Osteoporosis Risk Prediction

Executed project report with saved notebook outputs, tables, and charts

Project Description

Osteoporosis Risk Prediction

Project Summary

This project is a clean, student-friendly implementation of a osteoporosis risk prediction. The main goal is to identify osteoporosis risk from health and demographic factors. The notebook keeps the workflow simple: load the dataset, understand the columns, clean the data, train a baseline model, and check how well the model performs.

No external repository links or profile links are included in this description. Everything needed to understand and run the project is kept inside this folder.

Problem Statement

A raw dataset is useful only when it is converted into a clear decision-making workflow. In this project, the data is processed step by step so that important patterns can be found and a machine learning model can be trained. The expected output is based on Osteoporosis.

Files Included

File	Purpose	Quick Note
osteoporosis.csv	Main data source for this folder	1958 sampled rows x 16 columns
Notebook file	Complete Python workflow	Includes loading, cleaning, training, and evaluation
PDF report	Human-readable project summary	Matches the updated project structure

Important Columns

Column	Role
Id	Input feature used in the modelling workflow
Age	Input feature used in the modelling workflow
Gender	Input feature used in the modelling workflow
Hormonal Changes	Input feature used in the modelling workflow
Family History	Input feature used in the modelling workflow
Race/Ethnicity	Input feature used in the modelling workflow
Body Weight	Input feature used in the modelling workflow
Calcium Intake	Input feature used in the modelling workflow
Vitamin D Intake	Input feature used in the modelling workflow
Physical Activity	Input feature used in the modelling workflow
Smoking	Input feature used in the modelling workflow
Alcohol Consumption	Input feature used in the modelling workflow

Workflow Followed

1. Load the data from the local dataset file present in the same project folder.
2. Inspect the structure using shape, column names, missing values, and basic statistics.
3. Clean the dataset by removing empty columns, handling missing values, and keeping only useful features.
4. Prepare the model input by separating features and the target column.
5. Train a baseline model using a simple scikit-learn pipeline with preprocessing.
6. Evaluate the result using practical metrics such as accuracy, classification report, MAE, RMSE, or R2 score depending on the target type.
7. Write observations in a short and direct way so the project can be explained easily.

How to Run

1. Open the notebook in Jupyter Notebook, JupyterLab, Google Colab, or VS Code.
2. Keep the dataset files in the same folder as the notebook.
3. Run the cells from top to bottom.
4. Read the printed metrics and notes at the end before making conclusions.

Final Note

This version keeps the logic understandable and avoids unnecessary complexity. The aim is not to make the notebook look overloaded, but to make it readable, explainable, and useful for a student-level data science portfolio.

Executed Notebook Outputs

The outputs below were generated from the saved notebook after running the code cells in this project folder.

Cell 2 - 2. Load the dataset

```
Dataset shape: (1958, 16)
```

```
Id Age Gender Hormonal Changes Family History Race/Ethnicity \
0 104866 70 Female Normal Yes Asian
1 101999 32 Female Normal Yes Asian
2 106567 89 Female Postmenopausal No Caucasian
3 102316 77 Female Normal No Caucasian
4 101944 38 Male Postmenopausal Yes African American

Body Weight Calcium Intake Vitamin D Intake Physical Activity Smoking \
0 Underweight Low Sufficient Sedentary Yes
1 Underweight Low Sufficient Sedentary No
2 Normal Adequate Sufficient Active No
3 Underweight Adequate Insufficient Sedentary Yes
4 Normal Low Sufficient Active Yes

Alcohol Consumption Medical Conditions Medications Prior Fractures \
0 Moderate Rheumatoid Arthritis Corticosteroids Yes
1 NaN NaN NaN Yes
2 Moderate Hyperthyroidism Corticosteroids No
3 NaN Rheumatoid Arthritis Corticosteroids No
4 NaN Rheumatoid Arthritis NaN Yes

Osteoporosis
0 1
1 1
2 1
3 1
4 1
```

Cell 3 - 3. Understand the data

```
Rows: 1958
Columns: 16
```

```
column dtype missing_values unique_values
0 Id int64 0 1749
1 Age int64 0 75
2 Gender object 0 2
3 Hormonal Changes object 0 2
4 Family History object 0 2
5 Race/Ethnicity object 0 3
6 Body Weight object 0 2
7 Calcium Intake object 0 2
8 Vitamin D Intake object 0 2
9 Physical Activity object 0 2
10 Smoking object 0 2
11 Alcohol Consumption object 988 1
12 Medical Conditions object 647 2
13 Medications object 985 1
14 Prior Fractures object 0 2
15 Osteoporosis int64 0 2
```

Cell 4 - 4. Clean the data

```
Shape after simple cleaning: (1958, 16)
```

```
Id Age Gender Hormonal_Changes Family_History Race/Ethnicity \
0 104866 70 Female Normal Yes Asian
1 101999 32 Female Normal Yes Asian
2 106567 89 Female Postmenopausal No Caucasian
3 102316 77 Female Normal No Caucasian
4 101944 38 Male Postmenopausal Yes African American

Body_Weight Calcium_Intake Vitamin_D_Intake Physical_Activity Smoking \
0 Underweight Low Sufficient Sedentary Yes
1 Underweight Low Sufficient Sedentary No
2 Normal Adequate Sufficient Active No
3 Underweight Adequate Insufficient Sedentary Yes
4 Normal Low Sufficient Active Yes

Alcohol_Consumption Medical_Conditions Medications Prior_Fractures \
0 Moderate Rheumatoid Arthritis Corticosteroids Yes
1 NaN NaN NaN Yes
2 Moderate Hyperthyroidism Corticosteroids No
3 NaN Rheumatoid Arthritis Corticosteroids No
4 NaN Rheumatoid Arthritis NaN Yes

Osteoporosis
0 1
1 1
2 1
3 1
4 1
```

Cell 5 - 5. Select the target column

```
Selected target column: Osteoporosis
```

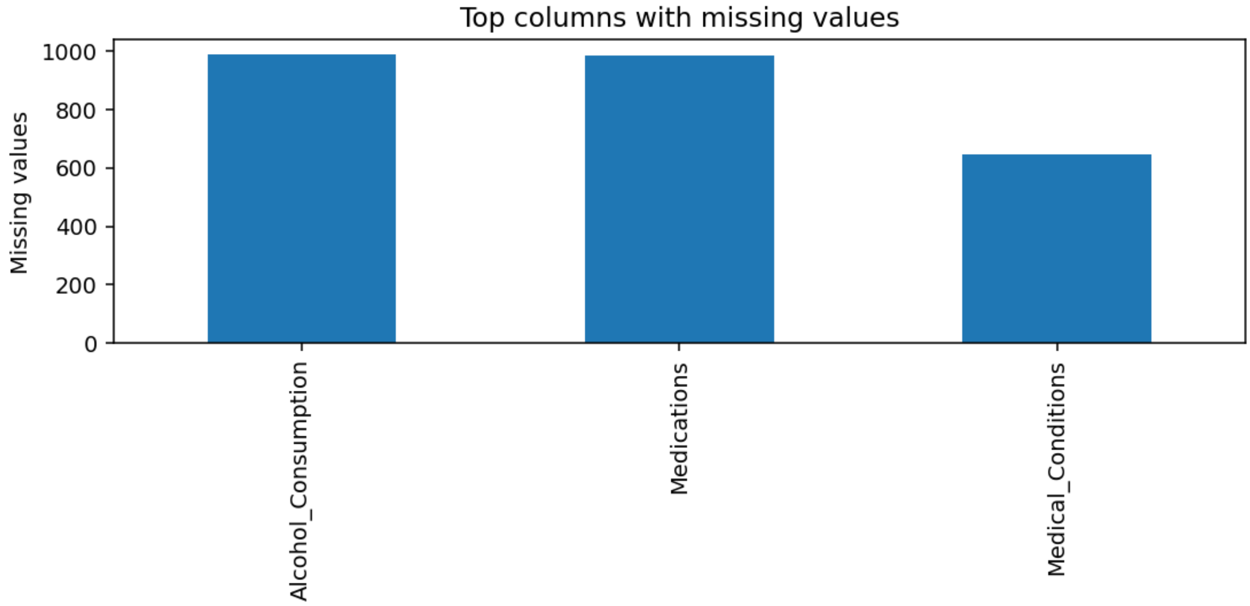
```
0 1
1 1
2 1
3 1
4 1
Name: Osteoporosis, dtype: int64
```

Cell 6 - 6. Quick EDA

```
Numeric columns: 3
Categorical columns: 13
```

```
count mean std min 25% \
Id 1958.0 105515.320735 2589.407806 101008.0 103348.5
Age 1958.0 39.070480 21.377406 17.0 21.0
Osteoporosis 1958.0 0.500000 0.500128 0.0 0.0

50% 75% max
Id 105469.0 107755.0 109996.0
Age 32.0 54.0 91.0
Osteoporosis 0.5 1.0 1.0
```



Cell 7 - 7. Prepare features and target

```
Dropped ID-like columns: ['Id']
Using 1200 sampled rows for model training speed.
Problem type: classification
Feature shape: (1200, 14)
Target unique values: 2
```

Cell 8 - 8. Train a baseline model

```
Training completed.
```

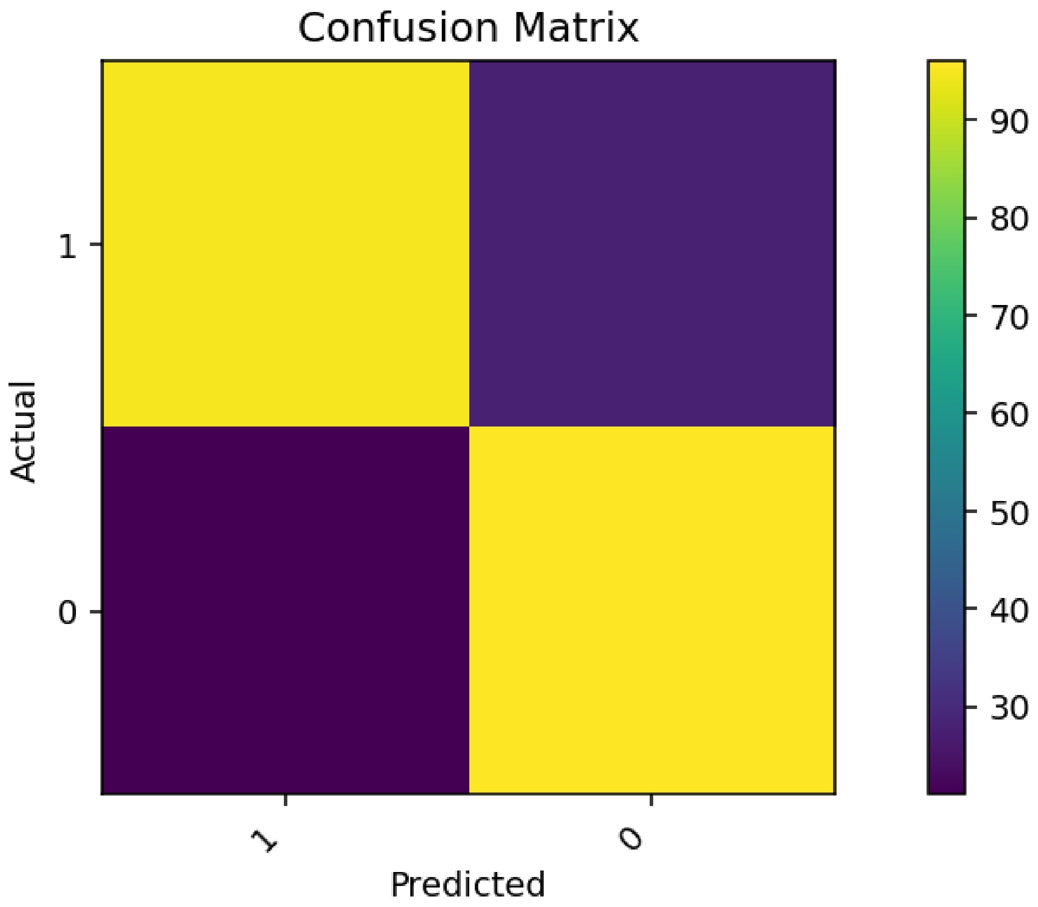
Cell 9 - 9. Evaluate the model

```
Accuracy: 0.7958

Classification report:
precision recall f1-score support

0 0.77 0.82 0.80 117
1 0.82 0.77 0.79 123

accuracy 0.80 240
macro avg 0.80 0.80 0.80 240
weighted avg 0.80 0.80 0.80 240
```


Confusion Matrix
Actual
1
0
Predicted
1
0
90
80
70
60
50
40
30